The bridge implication for the $(3, 4, \infty)$ lattice form

Restriction lemma for Ω_0

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Abstract

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This note records the lattice bridge already checked in `OmegaZero34/TetherInterface.lean`. The classification is `uniqueness_over_Q`. The bridge implication is `restriction_lemma`: non-degeneracy forces the scalar to be nonzero. It does not prove the geometric hypotheses, and it does not prove $X \simeq S^6$.

1 The form

On the rank-4 lattice with basis (γ, u, w, δ),

$$\Omega_0 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 6 & 0 \\ 0 & -6 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}.$$

Indices below are 0-based, so $\Omega_{0,03} = 1$ and the scalar in the Lean statement is the $(0, 3)$-entry.

2 Statements

Definition 1 (Invariant form)**.** An invariant rational form on this representation is a matrix $\omega \in \mathrm{Mat}_4(\mathbb{Q})$ satisfying

$$\begin{aligned} \omega^{\mathsf{T}} &= -\omega, \\ T_1^{\mathsf{T}} \omega T_1 &= \omega, \\ T_2^{\mathsf{T}} \omega T_2 &= \omega, \\ \det \omega &\neq 0. \end{aligned}$$

In the Lean development this is the structure `InvariantFormQ`.

Theorem 2 (Uniqueness over $\mathbb{Q}$)**.** *Let $\Psi \in \mathrm{Mat}_4(\mathbb{Q})$ satisfy $\Psi^{\mathsf{T}} = -\Psi$, $T_1^{\mathsf{T}} \Psi T_1 = \Psi$, and $T_2^{\mathsf{T}} \Psi T_2 = \Psi$. Then*

$$\Psi = \Psi_{03}\, \Omega_0.$$

The scalar Ψ_{03} may be zero. This is `uniqueness_over_Q`.

Theorem 3 (Bridge implication)**.** *Let $\omega \in \mathrm{Mat}_4(\mathbb{Q})$ be alternating, preserved by T_1 and T_2, and nondegenerate. Then*

$$\omega = \omega_{03}\,\Omega_0 \qquad \textit{and} \qquad \omega_{03} \neq 0.$$

Equivalently, $\omega = \lambda\Omega_0$ for $\lambda = \omega_{03} \in \mathbb{Q}^\times$. This is `restriction_lemma`*.*

Proof. By uniqueness over $\mathbb{Q}$, $\omega = \omega_{03}\,\Omega_0$. If $\omega_{03} = 0$, then $\omega = 0$, so $\det\omega = 0$, contradicting nondegeneracy. Thus $\lambda = \omega_{03} \in \mathbb{Q}^\times$. That is the only new step. □

3 What is not proved

The hypotheses are assumptions of the implication. This note does not prove a functorial alternating form on a family, monodromy invariance as a geometric theorem, or nondegeneracy from a construction. It does not prove holomorphic periods, existence of a compactification X, $\pi_1(X) = 1$, or $X \simeq S^6$. The monodromy here is the rank-4 group generated by T_1, T_2, T_0, not sextic-fourfold monodromy.